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American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science &Technology (FST)

Spring 21_22

Section: I

Group No: 6

Centralized Medical Application - CMA

A software Engineering project submitted
By

S/N	Student Name	Student ID	Contribution (%)	Individual Marks
07	Md Shahadot Hossain Shakib	20-42385-1	20%	
08	Shadril Hassan Shifat	20-42451-1	20%	
09	Abu Shaleh Md Kaium	20-42475-1	20%	
10	Gourob Kumar Das	20-42482-1	20%	
11	Ummee Zinat Refaiat	20-42487-1	20%	

The project will be Evaluated for the following Course Outcomes

Your Project will be Evaluated based on the following marking criteria	Total Marks
Requirements Analysis (functional, quality, and project requirements) [5Marks]	
System Design (UI/UX design) & Test case [5Marks]	
Project Management Planning [5Marks]	
Submission, Completeness, Spelling, Grammar and Organization [5Marks]	

Submission Date: 7 August, 2022

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Description of Student's Contribution in the Project work

Student Name: Md Shahadot Hossain Shakib
Student ID: 20-42385-1
Contribution in Percentage (%): 20
Contribution in the Project:

- Quality Attributes
- System test plan

_____
Signature of the Student

Student Name: Shadril Hassan Shifat
Student ID: 20-42451-1
Contribution in Percentage (%): 20
Contribution in the Project:

- System Features
- Project Requirements
- UI Design
- Project Scheduling

_____
_____
Signature of the Student

Student Name: Abu Shaleh Md Kaium
Student ID: 20-42475-1
Contribution in Percentage (%): 20
Contribution in the Project:

- System Test Plan
- Project Scheduling
- UI Design
- Quality Attributes

_____
Signature of the Student

Student Name: Gourob Kumar Das
Student ID: 20-42482-1
Contribution in Percentage (%): 20
Contribution in the Project:

- Risk Analysis
- System Test Plan

_____

Signature of the Student
Student Name: Ummee Zinat Refaiat Student ID: 20-42487-1 Contribution in Percentage (%): 20 <u>Contribution in the Project:</u> ▪ Risk Analysis ▪ System Test Plan _____ Signature of the Student

1. PRODUCT AND PROJECT DESCRIPTION

1.1 System Features

1. Software Registration

Functional Requirements:

1.1 Every time a user clicks "Register" on the homepage, the registration procedure must start.
1.2 In order to register, the user must give some personal information (Name, NID number, Date of Birth, Location, Cellphone number, Gender, Blood Group and Mail address) using the Sign-Up form.
1.3 Additionally, the user must create a secure password and re-enter it to validate password in the form.
1.4 Once the user has completed the required Manual Registration information, he/she needs to click 'Register' to initiate registration validation and completion.

Priority Level: High
Precondition: The user must accurately enter all required information and choose a secure password.
Cross-reference: 2.4, 4.1,4.2, 4.3, 4.4

2. Doctor Appointment

Functional Requirements:

2.1 The system will provide doctor recommendation to the user based on their disease prediction.
2.2 Additionally, a user can manually search for doctors by name.
2.3 The user will then be able to view the doctor's name, as well as their degree, specialty, and appointment time.
2.4 The user may book any doctor by clicking the 'Book' button next to the doctor's name.
2.5 Here, the user may also find out about various hospitals and their doctors.

Priority Level: High

Precondition: To book an appointment, the user must log in with their valid email and password.
Cross-reference: 1.1, 1.2, 1.3, 1.4, 3.4, 4.1,4.2, 4.3, 4.4

3. Disease Prediction
Functional Requirements:

3.1 User inputs such as blood pressure and pulse are required for disease prediction.
3.2 The user must check the box near the disease symptoms in the checkbox.
3.3 The user is required to provide information for the 'Duration' and 'Disease State' columns for any symptoms that have been checked in the checkbox.
3.4 Last, the user needs to click 'Submit' for disease prediction and after that he/she will be suggested doctors according to disease prediction.

Priority Level: High
Precondition: The user must accurately enter all required information.
Cross-reference: 2.3, 2.4, 2.5

4. Custom Exercise Guide to Patients
Functional Requirements:

4.1 In this option, the doctor may recommend exercises to the patient.
4.2 The doctor can provide the required documents or videos for the patient.
4.3 The patients are able to download or view the documents or videos.
4.4 Additionally, the user is able to communicate with the doctor in real time through live chat, and the doctor is able to monitor the patient's progress.
Priority Level: High
Precondition: The user must log in with their valid email and password and must have doctor's consultation subscription.
Cross-reference: 1.1, 1.2, 1.3, 1.4, 2.3, 2.4, 2.5

1.2 System Quality Attributes

QA 1- Availability: The system must be 99% available between 8.00 a.m. and 8:00 p.m. local time, and 97% available during the rest hours.

Priority Level: High
Precondition: The users must have enough internet connection.
Cross-reference: N/A

QA 2- Testability: Software should be able to detect when a system is at risk of failing. The maximum cyclomatic complexity should not be more than 15.

Priority Level: High

Precondition: N/A
Cross-reference: QA1**,** QA-4, QA-5

QA 3- Portability: The system shall run on the web-based platform. The user may access the system using a web browser on any device.

Priority Level: Medium
Precondition: N/A
Cross-reference: QA-2, QA-4, QA-5

QA 4 – Maintainability: It shouldn't take more than two hours for a maintenance programmer to make changes to an existing form. The maintenance programmers must be able to resolve any system issue in under three hours of manual effort in order to successfully fix the issue.
Priority Level: High
Precondition: The system should detect any errors.
Cross-reference: QA-1, QA-2, QA-5

QA 5–Flexibility: This system is going to be simple and straightforward to use. If anything has to be added or updated, a maintenance programmer may work on the software and produce a new version, including code modifications and testing, in less than 3-4 hours of effort.
Priority Level: High
Precondition: System should identify an error.
Cross-reference: QA-2, QA-3

1.3 Project Requirements

Time: It is estimated that it will take four months to finish the whole project.

Budget: In order to develop this system, we will need approximately 2.5 lakhs BDT as overall budget.

Human Resources: For the development of this system, we need a group of nine persons: one administrator, four software engineers, one software taster, one domain expert, and two members from the management team.

Internet Resources: A connection with a broadband speed of minimum 1 Mbps is required for the system to operate successfully.

Device Resources: Laptop, Desktop, Smartphone, Tablet.

Software Resources: HTML, CSS, Oracle Database, PHP, JavaScript.

User Interface Design: Pencil Desktop Application.

Environment: Adaptive Environment.

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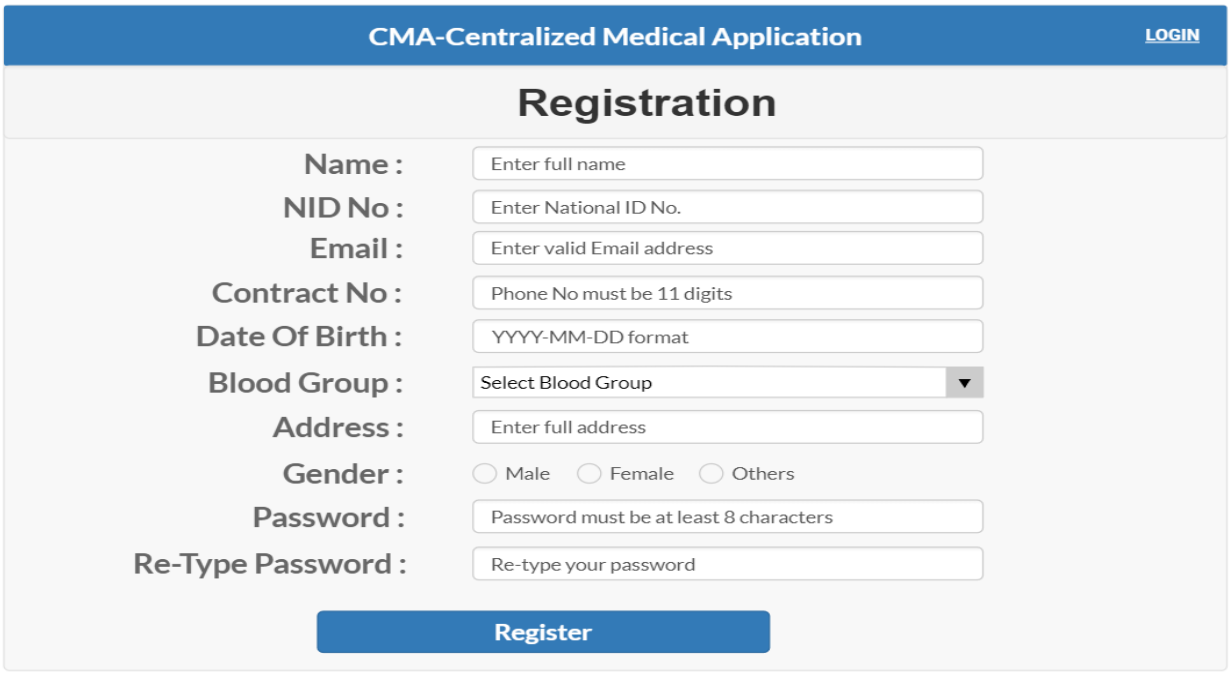
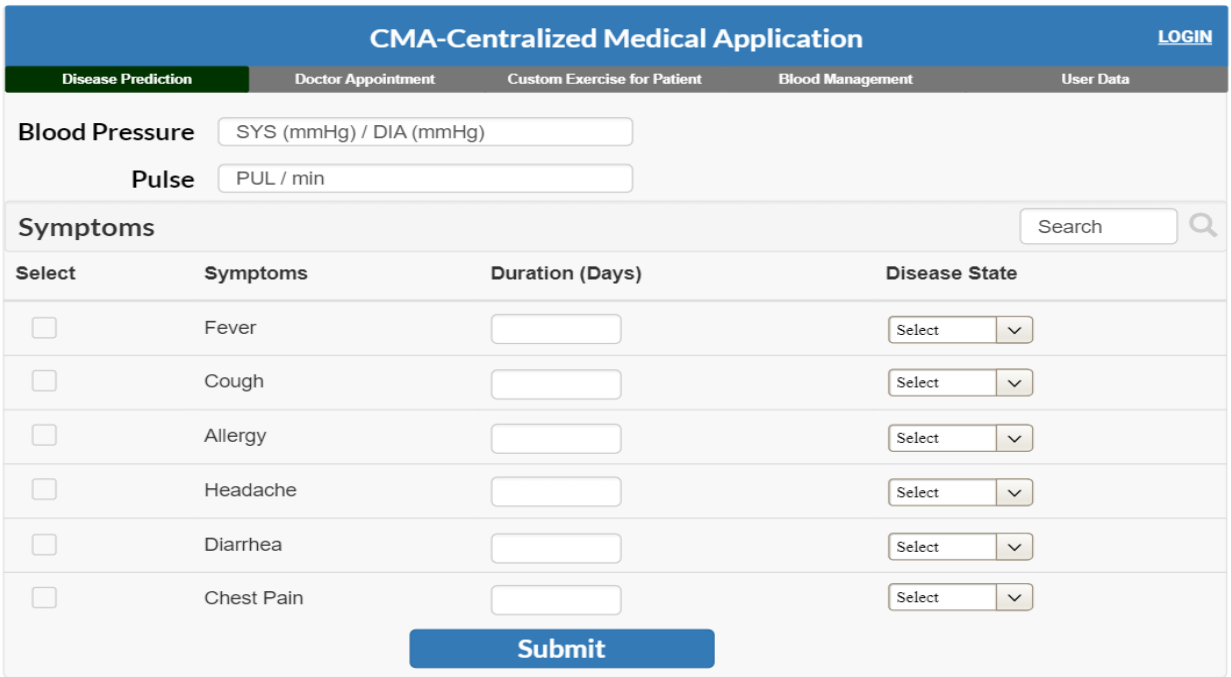
2. SYSTEM DESIGN SPECIFICATION

2.1 UI/UX Design





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CMA-Centralized Medical Application

LOGIN

Disease Prediction | Doctor Appointment | Custom Exercise for Patient | Blood Management | User Data

Doctor Schedule List

Search

Doctor Name	Education	Speciality	Hospital	Appointment Date	Appointment Time	Action
Dr. Shadril	MBBS MD	Nurologist	DMCH	2022-08-10	12.30-14.30	Book
Dr. Shifat	MBBS	Surgeon	MMCH	2022-08-09	13.40-15.50	Book
Dr. Hassan	MBBS	Surheon	Ibn Sina	2022-08-10	12.20-14.10	Book
Dr. Gourob	MBBS	Nurologist	Lab Aid	2022-08-11	14.30-16.30	Book
Dr. Prithy	MBBS MD	Surgeon	Appolo Hospital	2022-08-08	15.30-16.30	Book
Dr. Kaissa	MBBS	Nurologist	Al. Raji Hospital	2022-08-10	12.30-14.50	Book

CMH-Centralized Medical Hospital

LOGOUT

Disease Prediction | Doctor Appointment | Custom Exercise for Patient | Blood Management | User Data

Payment

Card | Mobile Banking | Online Banking

VISA

mastercard

AMERICAN EXPRESS

DBBL NEXUS

UnionPay 银联

Q cash

Pay 1000.00 BDT

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3. SYSTEM TEST PLAN

Project Name: Centralized Medical Application - CMA		Test Designed by: Abu Shaleh Md. Kaium		
Test Case ID: FR_2		Test Designed date: 06-Aug-22		
Test Priority: Medium		Test Executed by:		
Module Name: Doctor Appointment		Test Execution date:		
Test Title: Verify Doctor Appointment				
Description: Test web Doctor Appointment				
Precondition: The user needs to log in to the system.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to website and click on doctor appointment. 2. Click on search option. 3. Type a doctor's name or select from suggested doctors list. 4. Check doctors name, specialty, appointment date, hospital name, available time.	**For search option-** Doctor's name: Dr. Shadril Specialty : Neurologist Action: **Click on Book**	Go to payment option successfully.		

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5. Click on book option.				
Post Condition: This user's time slot will be reserved on database.				

Project Name: Centralized Medical Application -CMA		Test Designed by: Md Shahadot Hossain Shakib		
Test Case ID: FR_1		Test Designed date: 06-Apr-2022		
Test Priority : High		Test Executed by:		
Module Name: Disease prediction		Test Execution date:		
Test Title: Verify diseases prediction by suggesting doctor.				
Description: Test website disease prediction page.				
Precondition: The user must accurately enter all required information in the disease prediction page.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the diseases prediction option. 2. Enter Blood pressure and pulse. 3. Check option of symptoms. 4. Enter duration days. 5. Enter disease state. 6. Click on submit.	Blood pressure:120/80 Pulse: 70 Symptoms: Check Fever Duration: 7 Disease state: High	The interface should suggest some doctors for that disease.		

Project Name: Centralized Medical Application - CMA		Test Designed by: Abu Shaleh Md. Kaium		
Test Case ID: NFR_1		Test Designed date: 06-Aug-22		
Test Priority: High		Test Executed by:		
Module Name: Maintainability		Test Execution date:		
Test Title: Verify the responsiveness of System to solve problem within 3 hours				
Description: Test if system can solve the problem within 3 hours.				
Precondition: User must Login with valid username and password.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Go to the application and login. 2.Click on Doctor's appointment. 3. click on Book for taking appointment of a doctor.	**For search option-** Doctor's name: Dr. Shadril Speciality : Nurologist Action: **Click on Book**	Doctor's appoinment problem should be solved within 2 hours or less.		
Postcondition: N/A				

Project Name: Centralized Medical Application- CMA		Test Designed by: Abu Shaleh Md. Kaium		
Test Case ID: NFR_2		Test Designed date: 06-Apr-2022		
Test Priority: High		Test Executed by:		
Module Name: Availability		Test Execution date:		
Test Title: Verify the availability of the system				
Description: Test the availability of the system between 8:00 a.m. to 8:00 p.m.				
Precondition: User must login to the system				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website and click on disease prediction. 2. Use the system from 8:00 a.m. to 8:00 p.m. For 10000 times with automated software.	Blood pressure:120/80mmHg Pulse: 70 Symptoms: Fever Duration: 7 Disease state: Medium (Use different data on every test)	The system must be 99% available between 8.00 a.m. and 8:00 p.m. local time		

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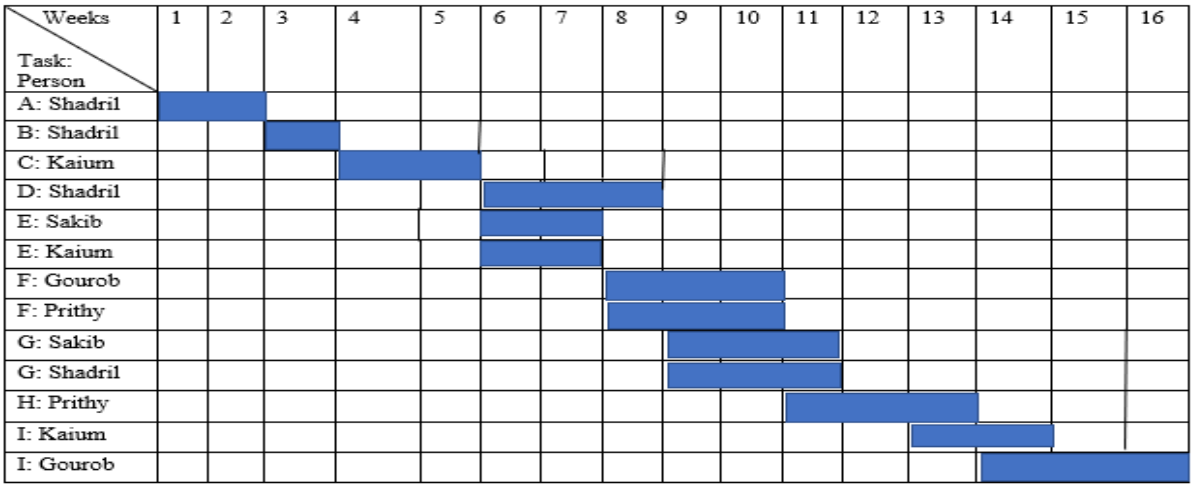
4. PROJECT MANAGEMENT PLAN

4.1 Project Scheduling



Activity Key:

A: Overall Design
B: Specify Login & Registration Module
C: Specify Doctor Appointment Module
D: Specify Disease Prediction Module
E: Code Login & Registration Module
F: Code Doctor Appointment Module
G: Code Disease Prediction Module
H: Integration Testing
I: System Testing

4.2 Risk Analysis

Risks	Category	Probability	Impact	RMMM
Size estimate may be significantly low	PS	50%	3	
Delivery deadline will be tightened	BU	60%	2	
Lack of Tools	DE	30%	3	
Staff inexperienced	ST	50%	3	
Less reuse than planned	PS	30%	3	
Technology will not meet expectations	TE	20%	4	
Change of requirements	PS	65%	1	

Impact values:

1- Catastrophic
2- Critical
3- Marginal
4-Negligible